Лабораторна робота №2

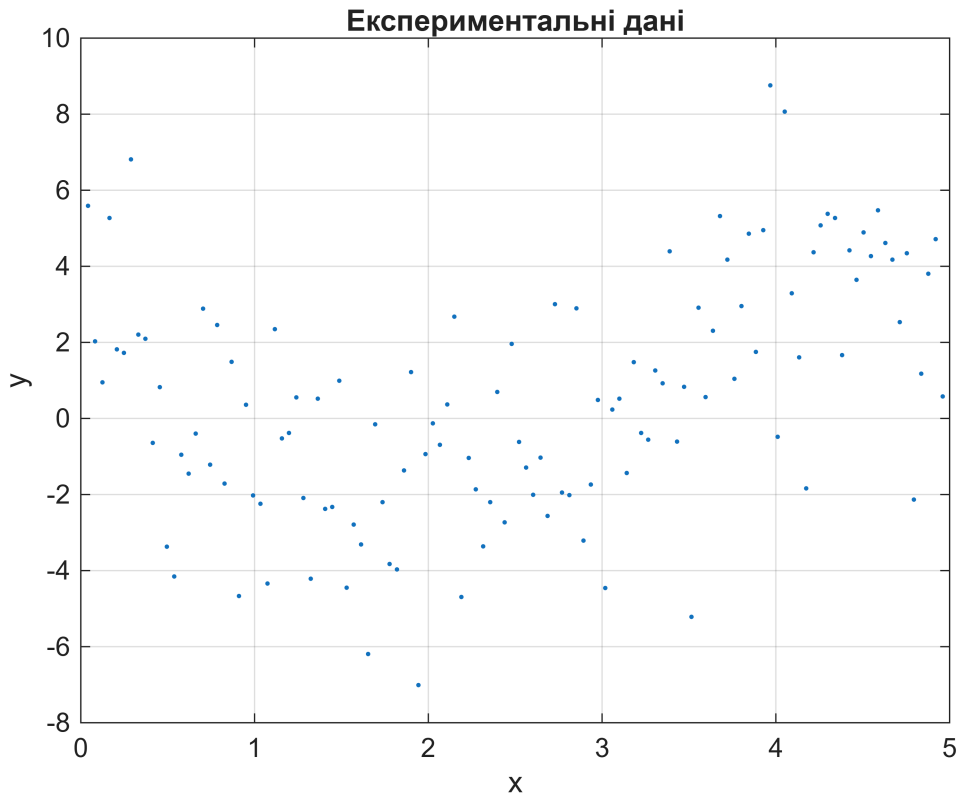
Яковець А., варіант 15

Апроксимація вибірки за допомогою М-функцій Curve Fitting Toolbox.

```
clear
clc
close all

fileName = 'v15.txt';
data = load(fileName);

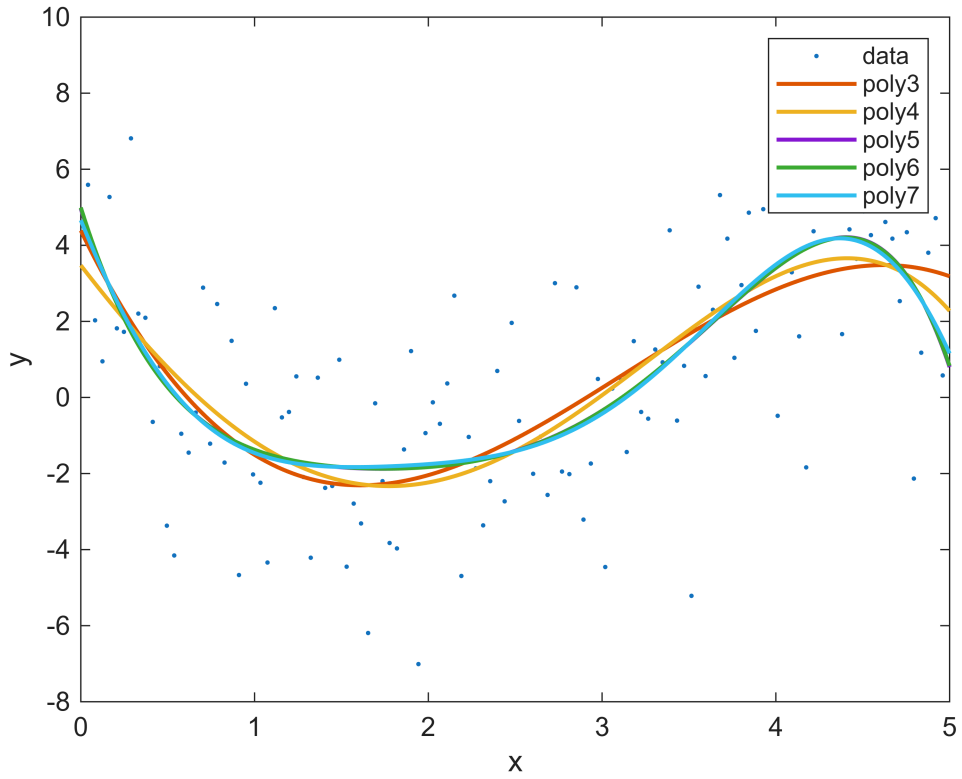
x = data(:,1);
y = data(:,2);
figure
plot (x, y, '.', 'MarkerSize',5)
xlabel('x')
ylabel('y')
title('Експериментальні дані')
grid on
```



Підбір поліному

```
[f3,gof3] = fit(x,y,'poly3');
[f4,gof4] = fit(x,y,'poly4');
[f5,gof5] = fit(x,y,'poly5');
```

```
[f6,gof6] = fit(x,y,'poly6');
[f7,gof7] = fit(x,y,'poly7');
plot (x, y, '.', 'MarkerSize',5)
hold on
plot(f3)
plot(f4)
plot(f5)
plot(f6)
plot(f7)
legend('data','poly3','poly4','poly5','poly6','poly7')
```



Порівняєм (R^2 має прямувати до 1, середньоквадратична помилка та сума квадратів помилок мають бути найменші)

```
gof3.rsquare
```

```
ans =
0.4483
```

```
gof4.rsquare
```

```
ans =
0.4563
```

```
gof5.rsquare
```

```
ans =
0.4721
```

```
gof6.rsquare
```

```
ans =
0.4721
```

```
gof7.rsquare
```

```
ans =
0.4726
```

```
gof3.rmse
```

```
ans =
2.3792
```

```
gof4.rmse
```

```
ans =
2.3722
```

```
gof5.rmse
```

```
ans =
2.3476
```

```
gof6.rmse
```

```
ans =
2.3580
```

```
gof7.rmse
```

```
ans =
2.3674
```

```
gof3.sse
```

```
ans =
656.6288
```

```
gof4.sse
```

```
ans =
647.1560
```

```
gof5.sse
```

```
ans =
628.2926
```

```
gof6.sse
```

```
ans =
628.2869
```

```
gof7.sse
```

```
ans =
627.7303
```

Найраще підходить поліном 5 степеня

```
ft = fittype('poly5');
opts = fitoptions('Method','LinearLeastSquares');
[fresult, gof, output] = fit(x,y,ft,opts);

fresult
```

```
fresult =
     Linear model Poly5:
     fresult(x) = p1*x^5 + p2*x^4 + p3*x^3 + p4*x^2 + p5*x + p6
     Coefficients (with 95% confidence bounds):
       p1 =     -0.1106  (-0.2289, 0.007826)
       p2 =       1.285  (-0.2027, 2.773)
       p3 =      -5.606  (-12.37, 1.155)
       p4 =       12.43  (-0.9703, 25.83)
       p5 =      -14.34  (-25.27, -3.413)
       p6 =       4.964  (2.219, 7.709)
```

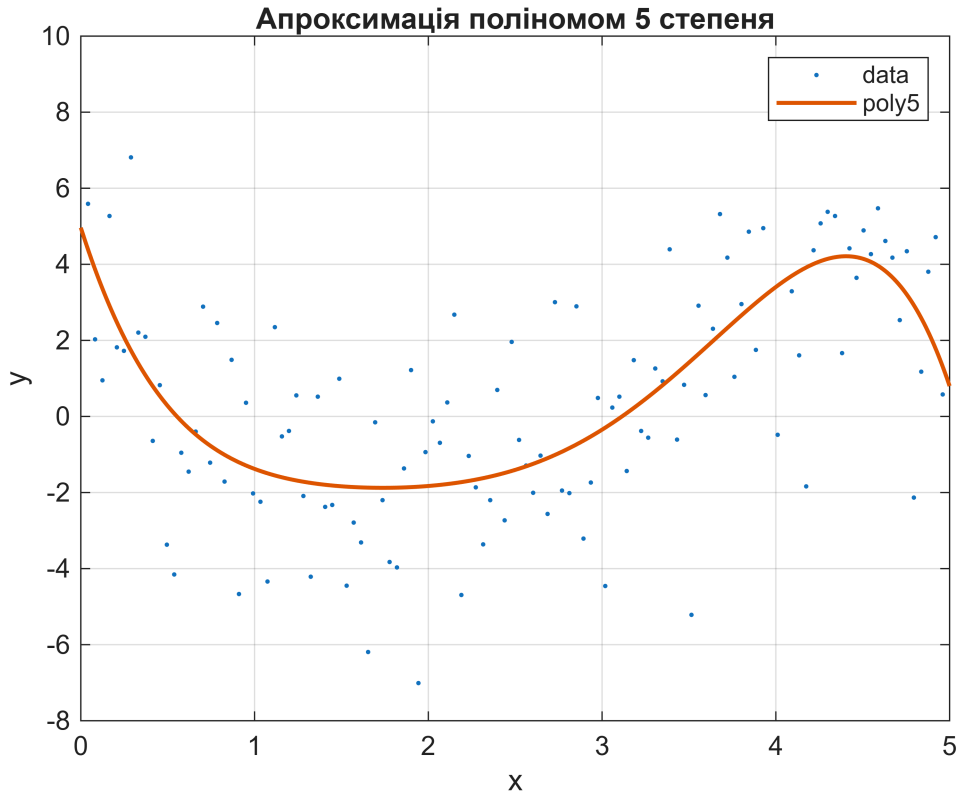
```
gof
```

```
gof =
struct with fields:
           sse: 628.2926
       rsquare: 0.4721
           dfe: 114
    adjrsquare: 0.4489
          rmse: 2.3476
```

```
output
```

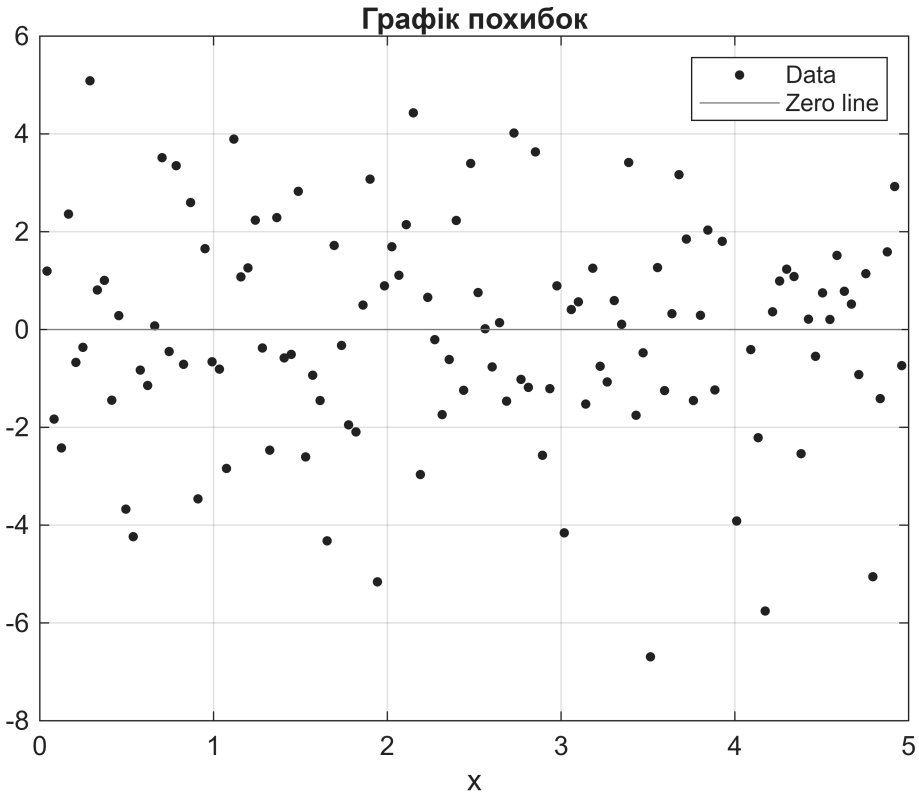
```
output =
struct with fields:
        numobs: 120
      numparam: 6
     residuals: [120×1 double]
      Jacobian: [120×6 double]
      exitflag: 1
     algorithm: 'QR factorization and solve'
    iterations: 1
```

```
figure
plot(x, y, '.', 'MarkerSize',5)
hold on
plot(f5)
grid on
legend('data','poly5')
title('Апроксимація поліномом 5 степеня')
xlabel('x')
ylabel('y')
```



Графік похибок

```
figure
plot(fresult,x,y,'residuals')
title('Графік похибок')
grid on
```




```
residual = output.residuals;
```

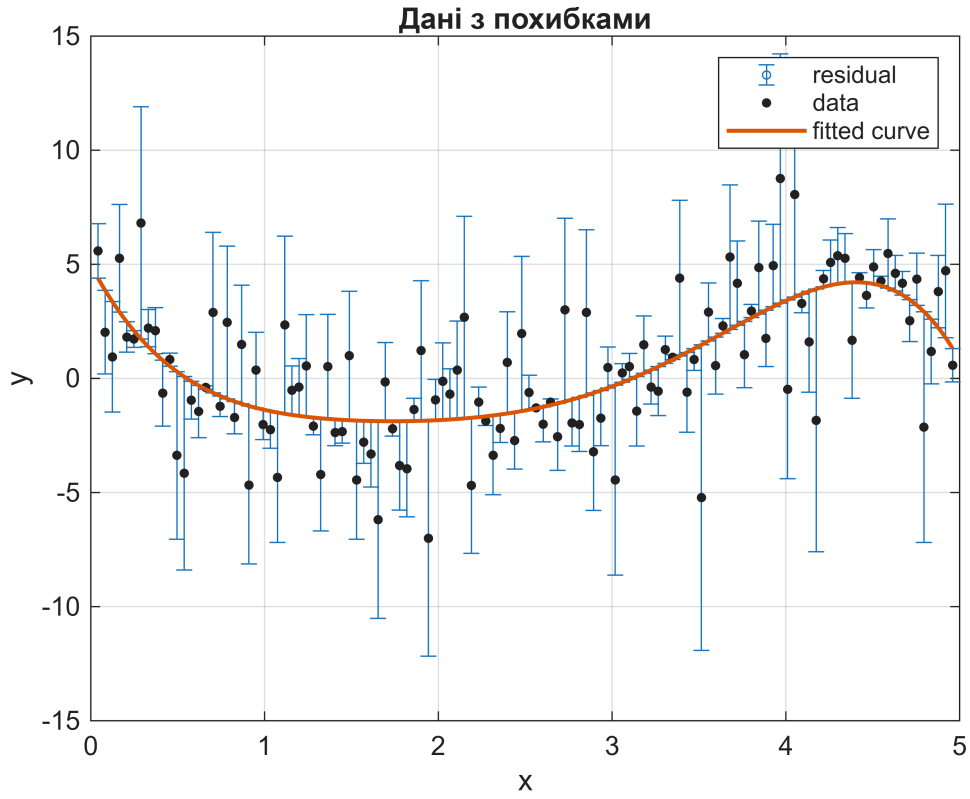
Графік похибок для кожної точки

```
u = abs(residual);

figure
errorbar(x,y,u,'o','MarkerSize',3)

hold on
plot(fresult, x, y)

grid on
legend('residual','data','fitted curve')
title('Дані з похибками')
```



Коеф полінома

```
coef = coeffvalues(fresult)
```

```
coef = 1×6
    -0.1106       1.2849      -5.6060      12.4290     -14.3403
```

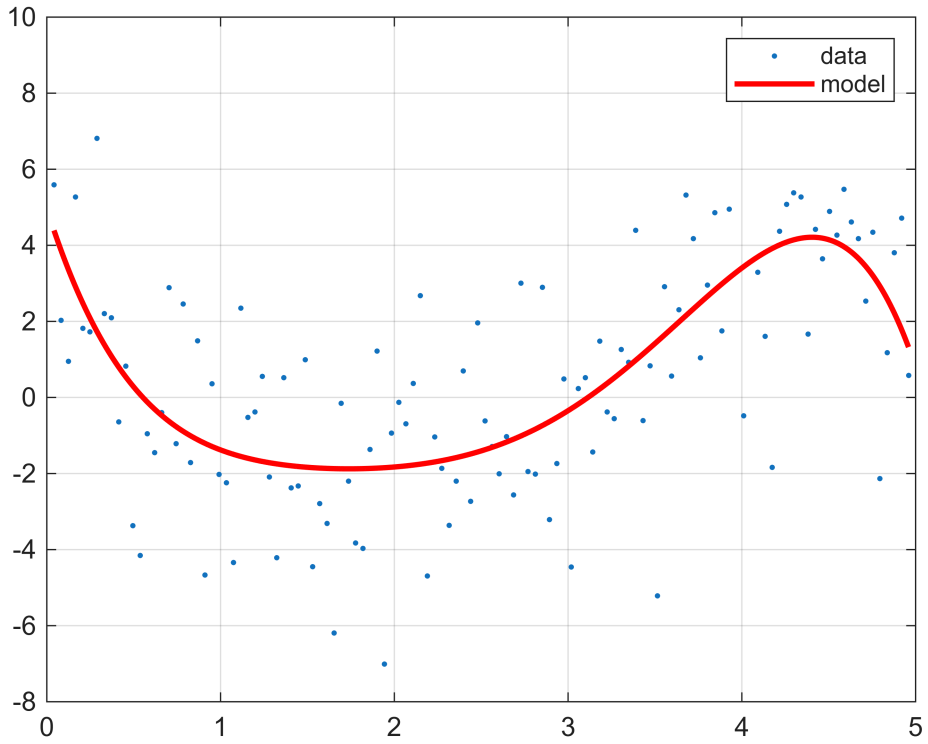
Гладший графік

```
xx = x(1):0.001:x(end);

ff = feval(fresult,xx);

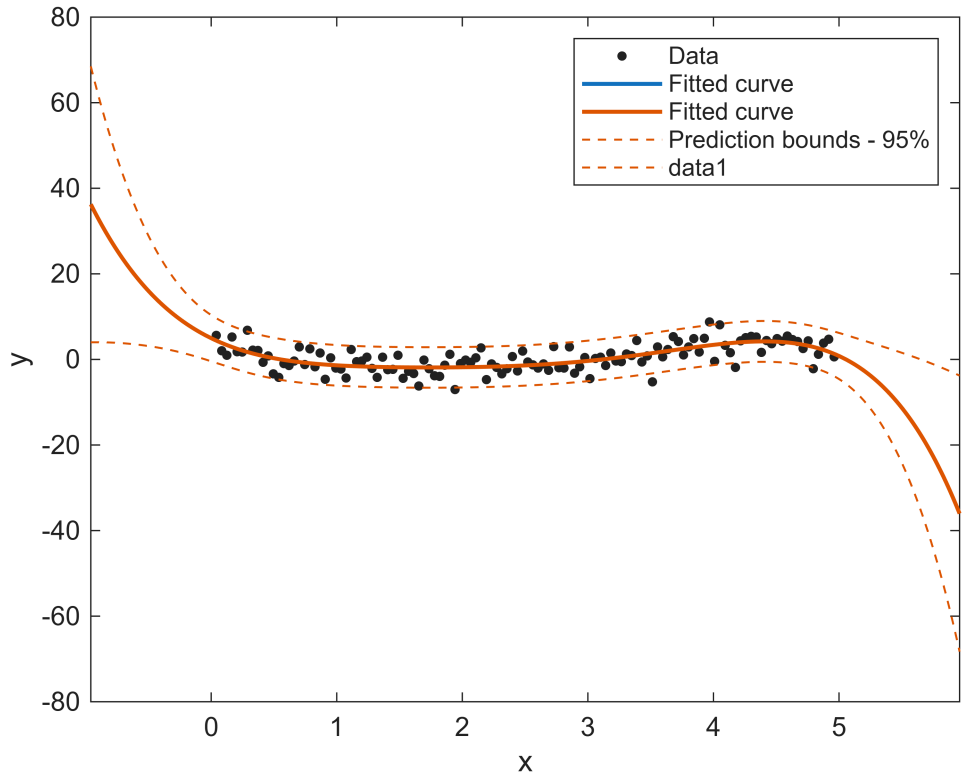
figure
plot(x,y,'.')
hold on
plot(xx,ff,'r','LineWidth',2)

grid on
legend('data','model')
```



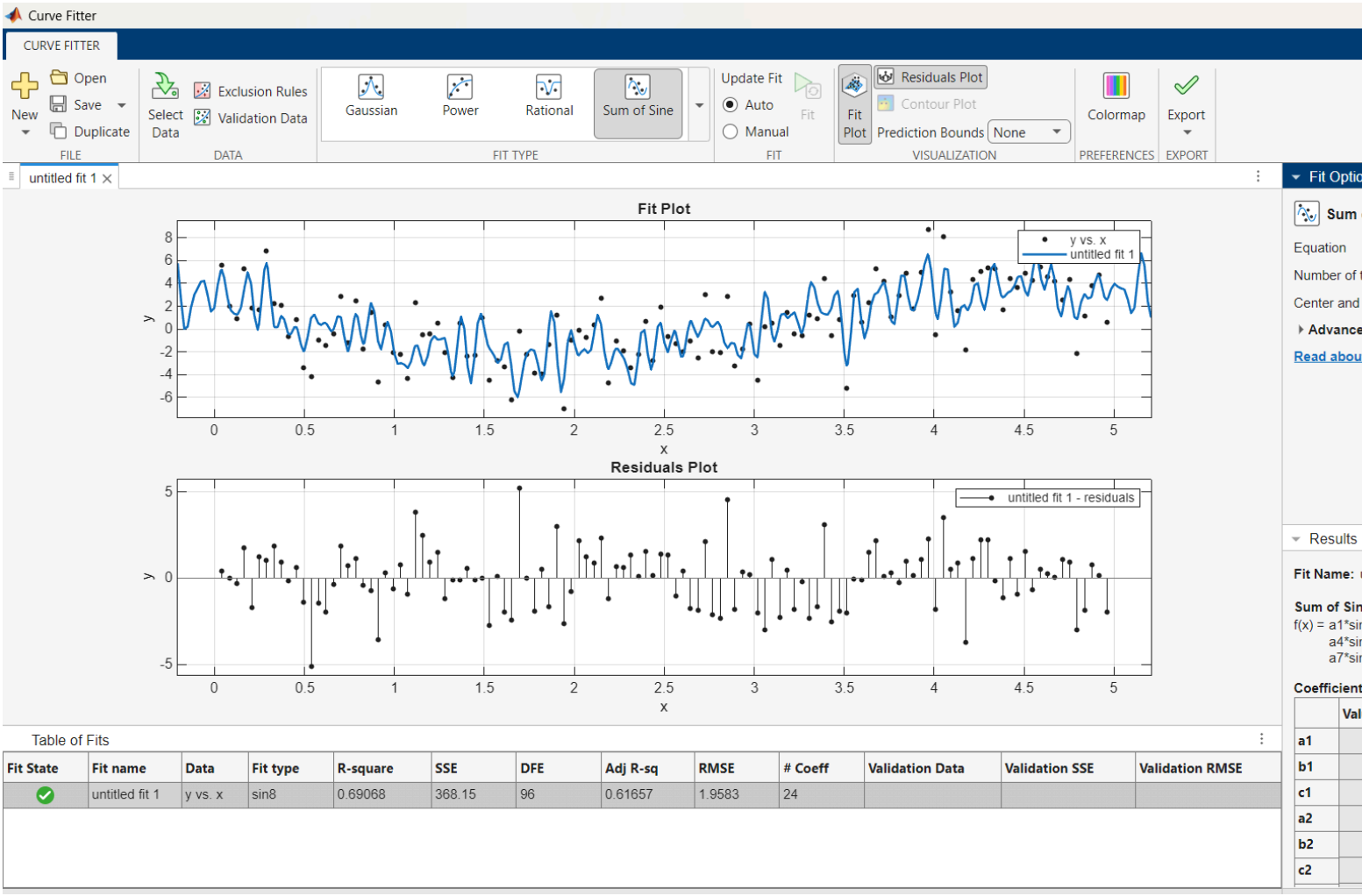
Екстраполяція

```
figure
plot(fresult,x,y)
xlim([min(x)-1 max(x)+1])
hold on
plot(fresult,'predobs')
```



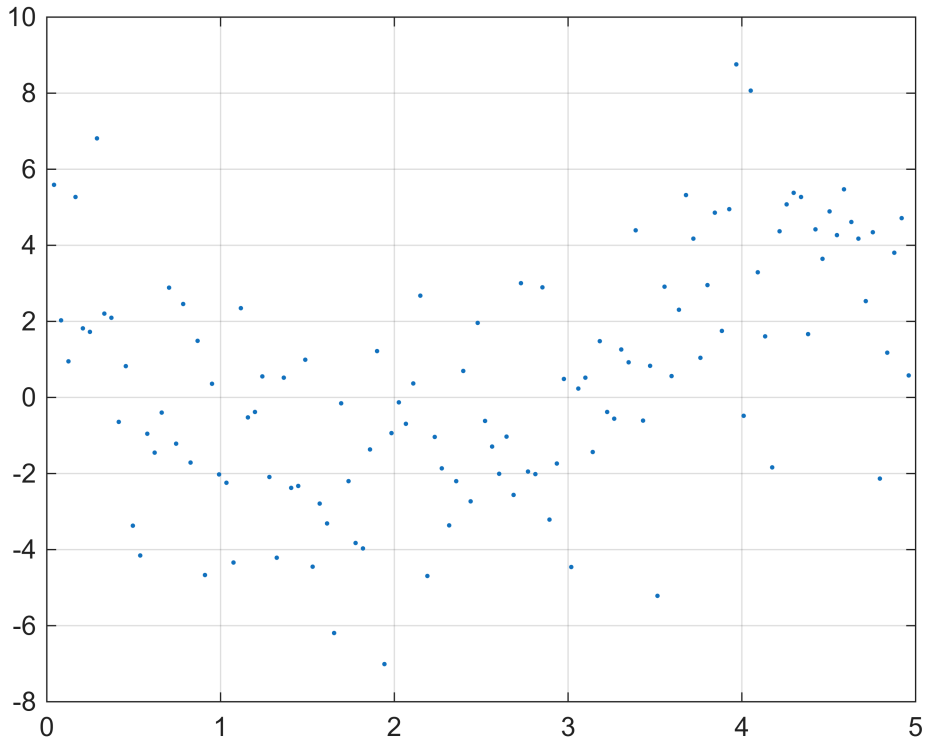
Апроксимація вибірки за допомогою користувацьких графічних інтерфейсів Curve Fitting Toolbox.

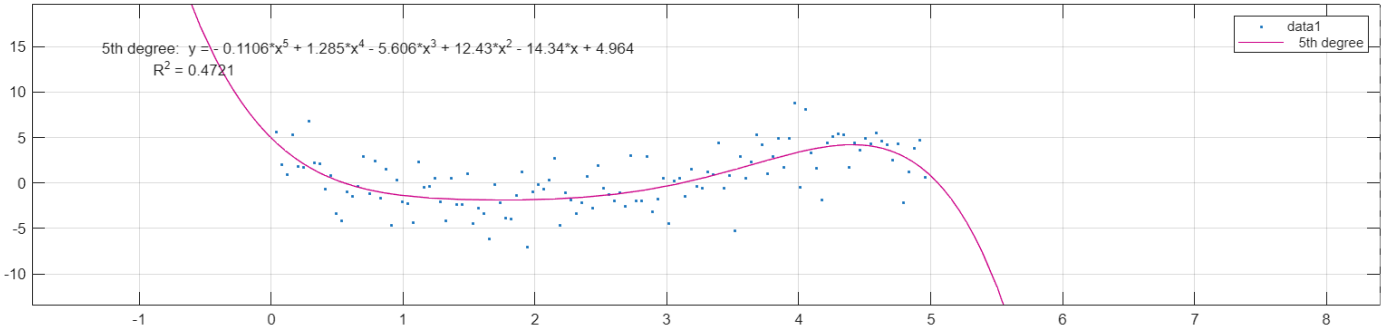
```
cftool
```

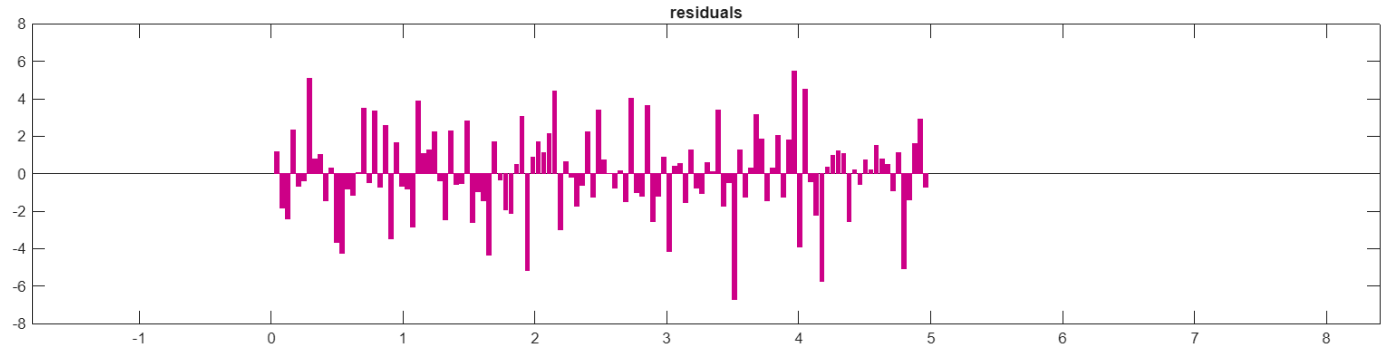


Апроксимація вибірки у графічному вікні Property Editor.

```
figure
plot(x,y,'.','MarkerSize',5)
grid on
```




5th degree: y = - 0.1106*x^5 + 1.285*x^4 - 5.606*x^3 + 12.43*x^2 - 14.34*x + 4.964
R^2 = 0.4721
data1
5th degree


residuals